Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION June-2025

NAME OF THE COURSE : M.Tech (Structural Engg.)
NAME OF THE PAPER : Advance Concrete Design
TIME: 3 Hours

SEMESTER: II
PAPER CODE: MSE 202
MAX MARKS: 100

Note:-

i. All questions are compulsory.
ii. Answer any two sub questions among a, b & c in each main question, each carrying 10 marks.

Q1 (2X10=20 marks)

(a) Explain guide lines to locate the yield line patterns based on the boundary conditions? (CO 1)

(b) Derive expression for minimum value of collapse Load for one-way slab using virtual work method? (CO 1)

(c) The factored moments capacities of one-way slab are Mi = 15KNm, Mj=25KNm. and Mo=18KNm. If the span of the slab is 5m, Determine the location of the plastic hinges and the collapse load? (CO 1)

Q2 (2X10=20 marks)

(a) Write the advantages and disadvantages of flat slabs? (CO 2)

(b) Design an interior panel of a flat slab with panel size 7 X 7 m supported by columns of size 400X400 mm. Provide suitable drop. Take live load as 3kN/m2. Use M20 concrete and Fe 415 steel. (CO 2)

(c) Write short notes on types and typical proportioning philosophy of the flat slabs. as per IS code? (CO 2)

Q3 (2X10=20 marks)

(a) State and discuss the various design components of an overhead water tank? Discuss the reason for preferring the W.S method for Water Tank Design? (CO 3)

(b) Design roof dome and the top ring beam for an elevated circular water tank of 14 m diameter and a total height of 3 meters, which is to be supported on a ring beam of 11m diameter. Assume that the ring beam is supported by 8 columns equally spaced. Used M25 grade concrete and Fe415 grade steel (assume the necessary data) (CO 3)

(c) Draw and identify the typical components of an Iintze tank. Discuss the economical proportioning and design principles of an intz Tank? (CO 3)

Q4 (2X10=20 marks)

(a) Write the various design steps involved in the preliminary design of a pre stressed flexural member?. (CO 4)

(b) Derive necessary equations and explain, how the limiting zone for the prestressing force is determined? (CO 4)

(c) Design a simply supported type-2 prestressed beam with M_{Total}=435kN-m. (including an estimated $M_{Self\ Wt}$ = 55kNm, Assume necessary data suitably). (CO 4)

Height of the beam is restricted to 650mm.
Prestress at transfer f_{p0} = 1300N/mm^2. and the prestress at service f_{pe}=1000N/mm^2.
Based on the grade of the concrete the
Allowable compressive stresses are 12.5N/mm^2 at transfer and 11.0N/mm^2 at service.
Allowable tensile stresses are 2.1 N/mm^2 at transfer and 1.6 N/mm^2 at service.
9-wire strand with nominal diameter of 10 mm are used.

Q5 (2X10=20 marks)

(a) What are the advantages of continuous members? Explain the effect of prestressing on indeterminate structures? Explain various methods of achieving continuity in prestressing member? (CO 5)

(b) State the assumptions made for analysis of secondary moments in continuous prestressed members?.Explain how the the secondary moments of a continuous prestressed member is analysed using theorem of three moments? (CO 5)

(c) A Continuous pre-stressed concrete beam ABC (AB=BC=10m) has a uniform rectangular cross-section with a width of 240mm and depth of 450mm. The cable carrying an effective force of 500kN is parallel to the axis of the beam and located at 120mm from the soffit. Determine the secondary and resultant moment at the central support B? (CO 5)

Table: For general LSM Design Use

Stress-Strain Values for			Fy=	500
K	Strain	Ultimate Stress - Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	400.0000	347.83	0.001839
0.8500	0.0003	425.0000	369.57	0.002148
0.9000	0.0007	450.0000	391.30	0.002657
0.9500	0.0010	475.0000	413.04	0.003065
0.9750	0.0020	487.5000	423.91	0.004120
1.0000	0.0035	500.0000	434.78	0.005674

Stress-Strain Values for			Fy=	415
K	Strain	Ultimate Stress-Strain Curve	Idealized Curve	Corresponding Strain
0.0000	0.0000	0.0000	0.00	0.000000
0.8000	0.0001	332.0000	288.70	0.001543
0.8500	0.0003	352.7500	306.74	0.001834
0.9000	0.0007	373.5000	324.78	0.002324
0.9500	0.0010	394.2500	342.83	0.002714
0.9750	0.0020	404.6250	351.85	0.003759
1.0000	0.0035	415.0000	360.87	0.005304